THE INJECTIVITY SITUATION REDUCES TO LOCALIZATIONS OF POLYNOMIAL ALGEBRAS WITH FINITELY PRESENTED MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0AT2 (sources/stacks-flat/pinned-original-source.tex line 7045).

Allowed standard prerequisites (editorial).

- Finite presentation over a subring from finite presentation over a quotient. Algebra, Lemma labelled algebra-lemma-finitely-presented-over-subring, cited at proof lines 7057-7059. Lets one replace B by the localization of a polynomial algebra.
- The situation and its conclusion. Same chapter, Situation labelled situation-mod-injective, invoked in the statement. Fixes the objects and the conclusion being reduced.
- Noetherianity of the special fibre. The fact that B/m_A B is essentially of finite type over a field, used at proof lines 7062-7066. Allows a finite choice of generators of K cutting out N modulo the maximal ideal.

Source reference. Original reference 0AT0: sources/stacks-flat/pinned-original-source.tex, source lines 7017–7017. Same-chapter pointer into the author environment situation at source line 7017; invoked at proof line 7054 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0561: sources/stacks-chow/tags.txt, source lines 6694–6694. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7059 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let A_0 be a local ring. If the lemma holds for every Situation 0AT0 with $A = A_0$, with B a localization of a polynomial algebra over A, and N of finite presentation over B, then the lemma holds for every Situation 0AT0 with $A = A_0$.*

Proof. Let $A \to B$, $u : N \to M$ be as in Situation 0AT0. Write $B = C/I$ where C is the localization of a polynomial algebra over A at a prime. If we can show that N is finitely presented as a C-module, then a fortiori this shows that N is finitely presented as a B-module (see Algebra, Lemma 0561). Hence we may assume that B is the localization of a polynomial algebra. Next, write $N = B^{\oplus n}/K$ for some submodule $K \subset B^{\oplus n}$. Since $B/\mathfrak{m}_A B$ is Noetherian (as it is essentially of finite type over a field), there exist finitely many elements $k_1, \ldots, k_s \in K$ such that for $K' = \sum Bk_i$ and $N' = B^{\oplus n}/K'$ the canonical surjection $N' \to N$ induces an isomorphism $N'/\mathfrak{m}_A N' \cong N/\mathfrak{m}_A N$. Now, if the lemma holds for the composition $u' : N' \to M$, then u' is injective, hence $N' = N$ and $u' = u$. Thus the lemma holds for the original situation. □